

ABCAD-QMS-002

Quality Definitions & Test Methods

agentic-bioinspired-cad: single-material, damage-tolerant, printable architectures

Document ID	ABCAD-QMS-002
Revision	B
Effective date	2026-09-25
Prepared by	Cameron B. Renteria
Owner / approver	Cameron B. Renteria
Companion documents	ABCAD-SOP-001 (loop operation)
Evidence index	results/README.md

1 Purpose and scope

This document is the controlled vocabulary and measurement specification for agentic-bioinspired-cad. It defines (i) every quality-relevant term used by the pipeline’s gates and instruments, (ii) the quantities measured and the equations behind them, (iii) the test methods (TM-1 ... TM-6) with their acceptance criteria and validity conditions, and (iv) the measured baseline recorded in `results/`.

The program goal is *damage-tolerant, 3D-printable architectures in a single material*, produced by a fully local agentic loop (LLM → Blender code-as-geometry → render → VLM critique → repair → manufacturability gate → STL). All function must come from geometry and topology — fiber entanglement, rotating weak planes, crack deflection — never from a hard/soft material contrast. Where this document and SOP-001 overlap, SOP-001 governs *operation*; this document governs *meaning, validity, and acceptance*.

2 Normative references

- ABCAD-SOP-001 Rev. B — *End-to-End Operation of the Generative Design & Print Loop*.
- Instruments:
 - printability (TM-2): `abccad/printing/print_audit.py`;
 - flaw-seeded retention (TM-3): `abccad/fea/damage_tolerance.py`;
 - validated linear FEA chain: `abccad/fea/voxel_plate_mesh.py` and `abccad/fea/run_tension.py`;
 - validated finite-strain solver: `abccad/fea/run_tension_finite.py`;
 - phase-field fracture and J-integral in MOOSE (TM-6): `abccad/fracture/`, with the decks in `abccad/fracture/decks/`.
- Science context (full references in `docs/LITERATURE.md`): woven metamaterials (Carton, Surjadi, Aymon, Portela); the helicoidal-composite review of Ning et al. (2024); printed-Bouligand pitch studies (15–30° per ply toughness band); imperfection-toughening studies of architected lattices (arXiv:2504.08873, arXiv:2507.13806).

3 Terms and definitions

3.1 Geometry and mesh integrity

STL

Triangulated boundary representation used for printing and voxelization. Units are millimetres throughout.

Welded vertex set

Vertices identified after rounding coordinates to 10^{-5} mm; the basis for all topological checks (far below any feature size, far above float noise).

Boundary (open) edge

An edge referenced by exactly one triangle. A tube with open ends contributes one 12- or 24-gon boundary loop per end.

Non-manifold edge

An edge referenced by more than two triangles. Bevel *miter folds* (Blender curve-bevel self-folding at path kinks) are the dominant source; slicers union them, but they invalidate topological centerline recovery.

Watertight

Zero boundary edges *and* zero non-manifold edges (pyvista feature-edge counts). Required for image-stencil voxelization to be trustworthy.

Euler characteristic χ

$\chi = V - E + F$ per connected component; a closed manifold tube (torus) gives $\chi = 0$, a capped cylinder $\chi = 2$. Values far outside $\{0, 2\}$ diagnose fold/degenerate contamination (for example, legacy bevel-swept woven coils measure $\chi = 24$).

Edge-connected component

Maximal set of triangles connected through shared *edges*. Fused fibers that touch only at isolated points remain separate edge-components; this is the correct fiber-separation test (vertex connectivity is not).

3.2 Voxelization and FEA meshing

Image-stencil voxelization

Rasterization of a closed STL into a boolean occupancy grid (VTK polydata-to-stencil inside test). Voxel pitch is denoted h [mm]; the baseline protocol uses $h = 1.0$ and the resolved protocol $h = 0.5$.

Keep-largest clean

Retention of the largest connected occupancy component before meshing; disconnected fragments (failed welds, floaters) are discarded — and their disappearance is itself a validity signal (TM-3 validity).

Loading plates / overlap

Two full-cross-section plates (thickness 2.0 mm) added at the tension faces, intersecting the structure by the *overlap* (0.8 mm) so plates grip fibers before they end. Plates are immune to flaw seeding.

Hex mesh band

Validated solver envelope: with the direct solver, $\lesssim 15\text{--}20$ k hexahedra per solve on a 16 GB host. Near-solid parts that would exceed the band at $h = 1.0$ are run at $h = 1.25$ *with the deviation disclosed in the record*; retention ratios remain comparable because each part is normalized to its own intact solve. With the algebraic-multigrid solver (ABCAD_FEA_SOLVER=amg, lineage-exact against the direct solver) meshes above 100 k hexahedra are within the envelope.

A_0 , H_0

Nominal cross-section area and gauge length of the plated specimen, recorded in the mesh sidecar and used to convert reactions to effective stress/strain.

Voxel-weld rule

Two solids are guaranteed to weld in the grid only if they interpenetrate by ≥ 0.4 mm (giving a contact lens $\gtrsim 1.6$ mm wide at $h = 1.0$). Tangent (zero-thickness) contacts do *not* reliably weld; coupons must be designed to this rule.

Resolution dependence

Absolute E_{eff} and floor margins depend on h (the woven specimen measures 82.7 MPa at $h = 1.0$ and 47.7 MPa at $h = 0.5$ with the same plating recipe). Every ranking claim must cite its voxel size.

3.3 Printability audit terms (print_audit.py)

EDT

Euclidean distance transform of the occupancy grid; the basis of all thickness/clearance measurements.

Formability ladder, lost@d

Morphological opening at diameter d : the fraction of material that cannot be formed by a tool of diameter d . A part “forms at d ” if $\text{lost}@d \leq 1\%$. Ladder rungs: $d \in \{0.3, 0.5, 0.8, 1.2\}$ mm.

max_formable_d

The *largest* ladder rung the part passes — the certified minimum feature size.

Clearance / fusion ladder

Twice the EDT of void space within 3 mm of solid; gaps below ≈ 0.8 mm are flagged as expected local fiber fusion for FDM.

Enclosed voids

Interior void components with no path to outside air — a trapped-resin defect class (resin process); informational for FDM.

Overhang fraction

Bed-aware fraction of surface area steeper than 45° (faces within 0.5 mm of z_{min} count as bed contact). $\leq 15\%$ is treated as support-free for FDM.

Process profile

Printer-specific limits; the default FDM profile assumes a 0.4 mm nozzle (blocking thin-feature limit 0.8 mm) and 0.2 mm layers. Selected with ABCAD_PRINT_PROCESS.

Verdicts

PRINT (no blocking issues), **PRINT w/ fixes** (non-blocking remediations listed), **SCALE/REDESIGN** (blocking violation with computed advice).

Auto-scale factor k

Deterministic repair emitted by the deep gate on a thin-feature failure: $k = \lceil (d_{\text{limit}}/d_{\text{certified}}) \rceil_{0.1}$, sanity-capped ($k \leq 4$, must fit the build volume). The scaled variant is re-audited before being recorded as printable.

Trust floor

Audit measurements are meaningful only ≥ 2 voxels; on large parts the auto-capped grid ($\leq \sim 326^3$) sets which rungs are trustworthy.

3.4 Damage tolerance (damage_tolerance.py)

Effective modulus E_{eff}

Linear small-strain uniaxial proxy from the validated chain (voxelize $\rightarrow$ plate $\rightarrow$ hex mesh $\rightarrow$ SfePy solve):

$$E_{\text{eff}} = \frac{F/A_0}{\Delta L/H_0}, \quad \text{material model } E = 3000 \text{ MPa}, \nu = 0.40, \varepsilon = 1\%.$$

E_{eff} is a structural (architecture $\times$ material) stiffness, scale-invariant as a fraction of E .

Flaw model

Per damaged replicate: $N = 3$ spherical flaws of radius $r = 1.5$ mm carved into the *member* region (plates immune; flaw centers kept clear of the grip bond), followed by re-cleaning. Replicates = 3, seeds pinned (base seed 20260703) for reproducibility.

Flaw dose f

Fraction of member material removed by the flaws (reported per replicate).

Retention R

$R = E_{\text{eff}}^{\text{damaged}}/E_{\text{eff}}^{\text{intact}}$ on identically meshed geometry.

Proportional-loss floor

$R_{\text{floor}} = 1 - f$: the retention of a hypothetical structure in which every removed element carried exactly its volumetric share of load.

Floor margin m

$m = \min_k R_k - (1 - \bar{f})$, with $\bar{f}$ the mean dose over the replicates. *The ranking quantity.* Interpretation rule: $m > 0$ redundant / damage-tolerant; $m \approx 0$ continuum-proportional; $m < 0$ load-path-sensitive (flaws sever critical members).

Damage-tolerance score

$\min_k R_k$ across replicates (worst sampled flaw placement), always reported *with* its dose and margin.

Zero-point verification

A solid continuum control must measure $m \approx 0$. Verified with the helicoidal plate loaded in-plane: $m = +0.0005$ at $h = 1.25$ and $m = -0.0002$ at $h = 0.5$. This anchors the instrument: nonzero margins on architected parts are signal, not meshing artifact.

Linear-screen limitation (mandatory caveat)

R is a *stiffness* retention under material removal. It cannot see crack-path mechanisms — the crack twisting and deflection behind the 15–30° per ply Bouligand band and behind enamel decussation. Those are resolved by TM-6.

3.5 Fracture (abcad/fracture/)

Validation gate

The set of checks MOOSE must pass before any of its numbers enters a record: elasticity and units, J-integral path independence, and the closed-form K benchmark (TM-6).

Work to common displacement

$W = \int R du$ integrated to the smallest maximum displacement of the runs being compared, after removing repeated rows from failed time steps. Between specimens of different stiffness this ratio mixes stiffness with dissipation; it must be reported together with the initial-stiffness ratio and the energy to peak load.

Crack tortuosity

Crack path length divided by its projected span; 1 for a straight crack.

Crack drift

Final vertical offset of the crack from the notch plane, as a fraction of the specimen height (full model).

3.6 Agentic-loop quality assurance

Two-phase split

Phase 1 (code emitter) runs in a child process that exits before Phase 2 (render / critic / repair) loads any model — two large models never co-reside (16 GB memory safety).

Critic approval

The local critic (qwen3-v1:8b) returns a schema-constrained JSON verdict with `match_quality` $\in \{\text{good, partial, poor}\}$ and `physical_stability`; *approval* requires `match=good`. Operating rule: the Ollama daemon MUST run with `OLLAMA_CONTEXT_LENGTH=8192` (prompt ≈ 3.5 k tokens; the model's chat template enables thinking) — at the 4096 default the verdict truncates or vanishes.

Repair attempt / stall guard

Code-fixer attempts are capped; a designer iteration that reproduces a previous iteration is flagged as a stall and the run ends cleanly (no design rounds on failed critiques).

Deep gate

The full quantitative print audit, run once per critic-approved design on the actually exported STL — never inside the repair loop. Its verdict merges into the manufacturability record; thin-feature failures trigger auto-scale.

Convergence event

One uninterrupted run achieving: repair $\rightarrow$ approval $\rightarrow$ deep gate $\rightarrow$ (auto-scale if needed) $\rightarrow$ **PRINT**-certified artifact + complete manifest.

Run manifest

`run_manifest.json` written at the end of every run: prompt, verdicts, artifact paths (including scaled variants), counters — the record of the run.

Terminal-state truthfulness

The recorded final state must reflect the last good artifact and honest gate verdicts (`approved=false` runs end with their evidence, never a fabricated success).

Swap watchdog

Every heavy step runs under a monitor that samples swap usage and aborts the step if swap exceeds the configured line (5120 MB for the loop on the 16 GB reference host) rather than stalling the machine.

4 Test methods

TM-1 — Mesh integrity (watertight/manifold report)

- **Purpose:** qualify an STL for voxelization, audit, and printing.
- **Apparatus:** watertight report (pyvista feature edges), CAD environment.
- **Procedure:** weld, count boundary and non-manifold edges, record bbox and triangle count.
- **Acceptance:** watertight = True for new parts. Open-tube swept lattices (woven, enamel) are remeshed as solids (`abccad.printing.remesh_watertight`) before certification; legacy exceptions with documented open ends or miter folds are tolerated only where the record shows slicer-union equivalence, and are barred from centerline-based processing.

TM-2 — Printability audit (per process)

- **Purpose:** certify a part against a printer process profile.
- **Apparatus:** `abccad/printing/print_audit.py`, process profile (default FDM 0.4 mm).
- **Procedure:** voxelize (automatic pitch, trust floor noted), EDT ladders, overhang, enclosed voids; emit verdict + CSV/MD report.
- **Acceptance:** verdict **PRINT**; thin features $\text{lost}@0.8 \leq 1\%$; overhang $\leq 15\%$ for support-free claims; enclosed voids = 0 for resin claims.

TM-3 — Damage-tolerance screen (flaw-seeded stiffness retention)

- **Purpose:** rank architectures by graceful stiffness retention under material loss.
- **Apparatus:** `abccad/fea/damage_tolerance.py` (composes only validated pieces); SfePy solves via `run_tension.py`.
- **Procedure:** baseline protocol at $h = 1.0$ (near-solid parts: $h = 1.25$, disclosed); resolved protocol at $h = 0.5$ with the AMG solver (the ranking of record); plates 2.0 / overlap 0.8, 3 replicates $\times$ 3 flaws $r = 1.5$ mm, pinned seeds; report E_{eff} , R_k , f_k , floor, margin, score; JSON + MD per part.
- **Acceptance / validity conditions (ALL required):**
 1. $E_{\text{eff}} \leq 1.02 E$ (material bound; a violation — for example 84 GPa on a degenerate remnant — voids the run; enforced in code as TM3-V1),
 2. keep-largest retains $\geq 70\%$ of the voxelized material (otherwise the part is disconnected at that voxel size; enforced in code as TM3-V2),
 3. intact hex count consistent with part volume ($\pm 30\%$ of V/h^3 + plates),
 4. a valid flaw band exists (gauge $\gtrsim 8$ mm between grip clearances at $r = 1.5$; otherwise re-orient the part or reduce r , disclosed),
 5. coupon designs obey the voxel-weld rule (≥ 0.4 mm interpenetration),
 6. swap watchdog active; solves within the hex band of the chosen solver.
- **Records:** `damage_tolerance.json`, `report.md`, and per-solve `e_eff.json` per part (archived under `results/damage_tolerance/`).

TM-4 — Loop convergence run

- **Purpose:** end-to-end qualification of the agentic loop as a manufacturing front end.
- **Apparatus:** the design loop per SOP-001; Ollama at 8192 context; swap watchdog.
- **Procedure:** drive a known-failing emit (saved code with an execution error) through the full loop uninterrupted.
- **Acceptance:** convergence event as defined above, with a truthful manifest. Non-approval outcomes are *valid* runs if they end with an honest terminal state; they are re-attempted, not patched.

TM-5 — FEA coupon design rules (for TM-3 inputs)

- Features intended to carry architecture signal must be ≥ 2 voxels; sub-voxel features (e.g. 0.2 mm laminate plies at $h \geq 1.0$) vanish — such parts measure as solid controls and MUST be labeled as such in reports.
- Crossing members: ≥ 0.4 mm interpenetration (voxel-weld rule).
- Target intact mesh inside the hex band; verify with a dry voxel count before solving.
- Verify intended geometry parameters independently of the generator where cheap (e.g. per-ply PCA angle check on exported pitch coupons).

TM-6 — Phase-field fracture and J-integral (MOOSE)

- **Purpose:** measure the crack-path mechanisms that TM-3 cannot see — crack deflection by rod orientation, crack twisting through rotating plies, and the crack driving force.
- **Apparatus:** MOOSE combined application (`solid_mechanics`, `phase_field`, `contact`); decks in `abccad/fracture/decks/`; deck generators and analysis scripts in `abccad/fracture/`.
- **Validation gate (must pass before any record):** (1) solid bar in uniaxial tension in project units returns $E_{\text{eff}} = E$ (3000.00 MPa, 0.000% error); (2) J-integral rings agree to $< 0.1\%$; (3) single-edge-notch tension, isotropic, $a/W = 0.2$: J within $\pm 5\%$ of K_I^2/E' with $K_I = F(a/W) \sigma \sqrt{\pi a}$ (achieved: 3.077×10^{-3} vs 3.223×10^{-3} , -4.5%). A crack on a symmetry plane requires `symmetry_plane` in `DomainIntegral`, otherwise J is exactly $2\times$ low.
- **Procedure:** orientation sweep of a notched specimen (half model); J at initiation versus orientation; full-model deflection with a seeded internal crack and variational-inequality irreversibility; ply-strip Bouligand decks at the coupon pitches; 3D blocks with rotating ply slabs; frictional-contact woven cells and weak-interface fracture decks.
- **Acceptance / validity conditions:** validation gate passed on the same build; relative results in model units unless the material is calibrated; load or work ratios between specimens of different stiffness reported with the stiffness ratio and the energy to peak load; each run's solver convergence recorded; results archived under `results/fracture/` and `results/woven_cell/`.

5 Measured baseline

5.1 Damage-tolerance ranking (TM-3)

All values from pinned-seed runs (`results/damage_tolerance/`); margins computed from the unrounded records. Doses differ by part and are shown.

Resolved protocol, $h = 0.5$ mm, AMG solver (ranking of record).

Part (load direction)	E_{eff} [MPa]	score	$\min R$	dose $\bar{f}$	floor $1-\bar{f}$	margin m
Helicoidal plate, in-plane (solid control)	554.0	0.997	0.0030	0.9970	-0.0002	
Bouligand coupon $\Delta\theta = 10^\circ$	446.3	0.991	0.0036	0.9964	-0.0055	
Bouligand coupon $\Delta\theta = 20^\circ$	474.4	0.990	0.0032	0.9968	-0.0070	
Bouligand coupon $\Delta\theta = 30^\circ$	520.3	0.989	0.0031	0.9969	-0.0075	
Double-twist laminate, in-plane	492.8	0.989	0.0025	0.9975	-0.0088	
Enamel lattice $\times 4$, sigmoid twist	1120.5	0.985	0.0051	0.9949	-0.0098	
Enamel lattice $\times 4$, linear twist	1096.0	0.984	0.0047	0.9953	-0.0109	
Bouligand coupon $\Delta\theta = 15^\circ$	447.8	0.986	0.0035	0.9965	-0.0110	
Enamel lattice $\times 4$, accelerating twist	1115.9	0.984	0.0049	0.9951	-0.0111	
Woven cube, fused crossings	47.7	0.947	0.0070	0.9930	-0.0462	

Baseline protocol, $h = 1.0$ mm (1.25 mm for the near-solid plates).

Part (load direction)	E_{eff} [MPa]	score	$\min R$	dose $\bar{f}$	floor $1-\bar{f}$	margin m
Helicoidal plate, in-plane (solid control)	941.2	0.998	0.0030	0.9970	+0.0005	
Double-twist laminate, in-plane	896.9	0.994	0.0026	0.9974	−0.0030	
Double-twist laminate, through-thickness	732.0	0.988	0.0069	0.9931	−0.0050	
Bouligand coupon $\Delta\theta = 10^\circ$	845.2	0.984	0.0073	0.9927	−0.0086	
Bouligand coupon $\Delta\theta = 15^\circ$	812.9	0.979	0.0069	0.9931	−0.0141	
Bouligand coupon $\Delta\theta = 20^\circ$	819.7	0.975	0.0074	0.9926	−0.0173	
Bouligand coupon $\Delta\theta = 30^\circ$	842.0	0.972	0.0072	0.9928	−0.0211	
Woven cube, fused crossings	82.7	0.881	0.0116	0.9884	−0.1076	

Findings of record. (1) The instrument zero point is verified (the solid control sits at its floor at both resolutions). (2) The coarse proxy exaggerates the spread: margins span +0.0005 to -0.108 at $h = 1.0$ but only -0.0002 to -0.011 at $h = 0.5$ for everything except the woven cube. (3) The *fused* woven cube is the only genuine outlier ($m = -0.046$ at $h = 0.5$): voxel-welded crossings convert entanglement into brittle continuous load paths. (4) Enamel ranks with the solid controls and the Bouligand coupons, and the twist law and the Bouligand pitch make no meaningful TM-3 difference — their mechanisms act on the crack path (TM-6). (5) The laminate plates' 0.2 mm plies are sub-voxel at $h \geq 1.0$; they measure as solid-class controls there (TM-5 labeling rule applies).

5.2 Fracture (TM-6)

Validation gate passed (Section 4, TM-6). Recorded mechanisms: rod orientation steers a free crack (drift +0.005 at 0° and 90° , -0.47 at 30° , +0.48 at 60°); crack tortuosity through the ply architecture of the Bouligand coupons is 1.46 / 1.82 / 1.60 / 1.66 at $\Delta\theta = 10 / 15 / 20 / 30^\circ$ per ply, peaking inside the 15-30° band; J at crack initiation rises by 7% at 45° relative to 0° . In the orientation sweep the peak-load ($1.32\times$) and work-to-common-displacement ($1.67\times$) ratios at 45° track the $1.81\times$ higher initial stiffness of the rotated anisotropic material, while the energy to peak load is unchanged ($0.98\times$); the energy consequence of crack deflection therefore remains to be measured. Non-fused woven cells with frictional crossings retain exactly the geometric fraction $(N-1)/N$ of their stiffness when one of N warp fibers is cut (0.500 for 2×2 , 0.667 for 3×3); weak interfaces ($G_c/10$) lower the peak load of a notched specimen to $0.32\times$ that of the uniform control.

5.3 Print certification status (TM-2, default FDM 0.4)

Artifact	Status
woven_fea_small_x20_smooth.stl	PRINT, no blocking issues — the release candidate. 555,622 triangles, watertight, 0 non-manifold, lost@0.8 = 0.17%, max_formable 1.2 mm, enclosed voids 0, overhang 13.4%. Produced by voxel remeshing (0.25 mm) and windowed-sinc smoothing of the certified $\times 2.0$ variant.
woven_fea_small_x20.stl	PRINT (clean), superseded for release by the remeshed file; retained as lineage.
woven_fea_small_x15.stl	PRINT (marginal, 0.99% < 0.8 mm).
Loop artifact (auto-scaled $\times 1.6$)	PRINT; the first artifact certified end to end by the loop.

5.4 Loop qualification status (TM-4)

Convergence event witnessed (uninterrupted; swap peak 2236 MB): repair (2 attempts) $\rightarrow$ approval (match=good, stability=stable) $\rightarrow$ deep-gate honest thin-feature fail (14% < 0.8 mm, correct $\times 1.6$ advice) $\rightarrow$ auto-scale $\rightarrow$ re-audit **PRINT** $\rightarrow$ complete manifest (archived in **results/agent_runs/**). Critique reliability at 8192 context: zero dropouts across two consecutive full runs (six verdicts). Code-fixer record: eight consecutive runs at ≤ 3 attempts.

6 Records, traceability, change control

- Every TM-2, TM-3, and TM-6 run stores JSON/CSV/MD evidence; every loop run stores a manifest in its run folder; heavy-step logs are kept with the run.
- Records that back published numbers are archived under **results/** and checked by the gold tests (**tests/test_golds.py**).
- Seeds, voxel pitches, and any protocol deviation (e.g. $h = 1.25$ on near-solid parts, part re-orientation) **MUST** appear in the stored record.
- Changes to gates, instruments, or this vocabulary are recorded (what / why / files / verification) and bump this document's revision.

Revision history

Rev	Date	Change
A	2026-07-05	Initial release: vocabulary, TM-1 to TM-6, measured baseline.
B	2026-09-25	Public release: identifiers renamed to the abcad namespace; TM-6 defined as the MOOSE phase-field fracture and J-integral method with its validation gate; 0.5 mm resolved TM-3 ranking of record added with resolution-dependence rule; fracture baseline added.